DetectorAI at ArGuard Shared Tasks: Calibrated Multimodal and Label-Frequency-Aware Systems for Arabic Meme Safety Detection

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Abstract

We present our systems for ArGuard 2026 Track A, which aims to detect harmful content in Arabic memes. The track includes two tasks: a binary task for identifying hateful memes (A1) and a multi-label task for predicting specific categories (A2). For A1, our approach combines a fine-tuned Qwen3-VL sequence classification model, Arabic text encoders, and a Qwen text LoRA classifier. We then apply top-k calibration and make targeted adjustments based on model disagreements. On the official test set, our system reached a macro-F1 score of 0.7896 and an accuracy of 0.8200. For A2, we found that using thresholds from the development data caused us to overpredict common labels and miss rare ones. Our best A2 submission uses an ensemble of six Arabic text models, with label-frequency-aware top-k calibration and rare-label boosting. This approach achieved a macro-F1 of 0.3601 and a micro-F1 of 0.5150. Overall, we found that careful calibration of a few fine-tuned Arabic and multimodal classifiers contributed more to our results than using larger zero-shot vision-language models.

1 Introduction

Detecting harmful content in Arabic is challenging for both linguistic and multimodal reasons. Harmful memes may use dialectal Arabic, sarcasm, humor, cultural references, offensive visuals, and noisy overlaid text from OCR. ArGuard 2026 Track A (Alam et al., 2026) uses the AHA-Memes dataset (Kmainasi et al., 2026) and includes two subtasks: A1, which determines if an Arabic meme is hateful, and A2, which identifies one or more specific categories such as *Contempt*, *Mocking*, *Sarcasm*, or *Slurs*.

We explored three types of models: Arabic text encoders fine-tuned on OCR text, open vision-language models used in both zero-shot and fine-tuned settings, and calibrated ensembles that combined multimodal and text-only evidence. For A1, combining multimodal evidence from Qwen3-VL (Bai et al., 2025) with text classifiers was helpful, but the final macro-F1 depended a lot on which examples were predicted as hateful. For A2, the main challenge was per-label calibration. A multi-label ensemble with high recall achieved a good micro-F1, but improving macro-F1 required special attention to rare labels.

Our main contributions are: (1) Arabic OCR text is a strong signal for detecting hateful memes; (2) fine-tuned vision-language models help with A1 when used as calibrated probability sources instead of direct zero-shot labelers; (3) A2 performance improves with label-frequency-aware top-k post-processing, since the evaluation metric gives equal weight to rare and common labels; and (4) making a small, targeted adjustment to the most uncertain A1 examples improved our final ranking.

2 Task and Data

ArGuard Track A has been constructed using an Arabic multimodal meme dataset which is connected to recent Arabic harmful meme resources and to multimodal safety evaluation initiatives (Kmainasi et al., 2026; Alam et al., 2024; Kmainasi et al., 2025a,b). In each case, there is an image, the Arabic text obtained by OCR, a binary label for A1, and a list of fine-grained labels for A2. The official setup included 3,500 training examples, 500 labeled development examples, 500 blind development-test examples, and 500 blind final test examples, the latter of which was evaluated on CodaBench using hidden labels.

A1. The task involves binary classification with the labels *Hateful* and *Not Hateful*; the official metric is macro-F1, with accuracy and class-wise F1 scores provided for diagnostic purposes.

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Task	Models	Calibration
A1	Qwen3-VL-4B LoRA + MARBERTv2 / CAMeLBERT + Qwen text LoRA	Prob. blend, top-k hateful count, 3-ex. surgical swap (0.7896 Ma-F1)
A2	3 CAMeLBERT-mix + 3 MARBERTv2 seeds	Per-label top-k, rare-label boost, common-label restore, min-1 label (0.3601 Ma-F1)

Table 1: Final Track A pipelines.

A2. The task involves carrying out multi-label classification across ten categories: contempt, dehumanization, exclusion, humor, incitement, inferiority, mocking, other, sarcasm, and slurs; the official metric is macro-F1 with respect to the labels, and micro-F1, hamming loss, subset accuracy, and per-label F1 are given for diagnostic purposes.

3 Systems

Table 1 summarizes the two final pipelines. A1 uses both image and OCR-text evidence, while A2 relies on a calibrated Arabic text ensemble because our multimodal A2 attempts were less stable than label-wise text calibration.

3.1 Text Encoders

We fine-tuned Arabic transformer encoders on the OCR text. The strongest text-only family used MARBERTv2 (Abdul-Mageed et al., 2021) and CAMeLBERT-mix (Inoue et al., 2021) with multiple random seeds. We also tested AraBERT-style text baselines (Antoun et al., 2020), but they were not selected for the final submissions. For A1, each model produced a hateful probability and a development-tuned decision threshold. For A2, the same family was trained with sigmoid multi-label heads and class-wise threshold tuning. Text preprocessing used light Arabic normalization, avoiding aggressive normalization that could remove dialectal or offensive lexical cues.

3.2 Qwen Text LoRA and Vision-Language Models

For the text-based generative model, we fine-tuned Qwen3-4B (Yang et al., 2025) using LoRA (Hu et al., 2021) on the OCR text. For multimodal classification, we fine-tuned Qwen3-VL-4B (Bai et al., 2025) using MS-SWIFT (ModelScope Team, 2025) with the meme image together with its OCR text.

We also explored open vision-language models as alternative baselines. Fanar/Oryx (TEAM et al., 2026) was evaluated using zero-shot prompting, while MemeIntel (Kmainasi et al., 2025a) and MemeLens (Shahroor et al., 2026) were tested as meme-oriented alternatives. These models were less reliable than the fine-tuned Qwen3-VL and Arabic text models and were therefore not selected for the final system.

3.3 A1 Ensemble and Surgical Calibration

Our final A1 system used a probability blend of the Qwen3-VL-4B classifier and a text ensemble:

$$p = 0.60\, p_{\text{VLM}} + 0.40\, p_{\text{text}},$$

where p_{text} averaged six Arabic text encoders and the Qwen3 (Yang et al., 2025) text LoRA. Rather than a fixed threshold, we selected the top-k examples by blended score, which is useful under macro-F1 since the hateful-class precision–recall tradeoff depends strongly on the predicted positive count.

We selected the blend from a development sweep over VLM weights 0.55–0.65 in increments of 0.025, re-tuning the decision threshold for every weight. The resulting macro-F1 scores occupied a narrow 0.7550–0.7584 range; the 0.60 weight gave 0.7561 and was retained as a central, less endpoint-dependent setting. Its eventual selection over the neighboring 0.55 and 0.65 variants, as well as the final top-k operating point, was informed by aggregate CodaBench feedback.

A top-162 version reached 0.7849 macro-F1 (109 true positives). Since a single correct exchange between a false positive and a false negative can raise macro-F1 substantially, we created surgical swap variants that preserve the positive count but exchange a small number of borderline examples using an alternate Qwen text-heavy ranking. The best variant, `swap_qseq_heavy_n3`, achieved 0.7896 macro-F1.

3.4 A2 Ensemble and Label-Frequency Calibration

Direct thresholding transferred poorly from development to final test: our early high-recall ensemble over-predicted common labels (*Sarcasm*, *Humor*, *Mocking*, *Other*) while missing rare ones (*Contempt*, *Inferiority*), giving reasonable micro-F1 but weak macro-F1.

Our best A2 system averaged label probabilities across six Arabic text models (three

CAMeLBERT-mix, three MARBERTv2 seeds) with slight seed/family weighting, then selected per-label top-k predictions. Calibration was guided by CodaBench diagnostic support counts: rare labels were boosted while common labels were restored to preserve recall. The final file, `hybrid_restore_common_r03_min1`, enforces at least one predicted label per example and reached 0.3601 macro-F1.

4 Experimental Setup

Table 2 summarizes the training settings. The exact text checkpoints were `UBC-NLP/MARBERTv2` and `CAMeL-Lab/bert-base-arabic-camelbert-mix`; each family used seeds 13, 42, and 777 and a maximum length of 192. A1 optimized class-weighted cross-entropy, whereas A2 used binary cross-entropy with positive-class weights clipped at 20. All models used AdamW. Text encoders used weight decay 0.01, warmup ratio 0.08, and early-stopping patience 2. Qwen3-4B used maximum length 256, LoRA dropout 0.05, and the attention and MLP projection modules as targets. Qwen3-VL used maximum length 2048, warmup ratio 0.05, and all linear layers as LoRA targets. Qwen models and the text ensembles ran on one NVIDIA H200 141GB GPU; follow-up inference and calibration used one A100 80GB GPU. Thresholds and top-k rules were tuned on development data and refined using aggregate CodaBench feedback.

Model	Epochs	LR	Batch	LoRA r/α	Prec.
MARBERTv2/CAMeL (A1)	6	2e-5	16	–	fp32/fp16
MARBERTv2/CAMeL (A2)	8	2e-5	16	–	fp16
Qwen3-4B LoRA	4	1e-4	16†	16/32	bf16
Qwen3-VL-4B	5	1e-5	8	16/32	bf16

Table 2: Training hyperparameters. †Effective batch via gradient accumulation (2×8).

5 Results

Table 5 gives the A1 component ablation on the labeled development split. The 0.60/0.40 blend improves over both of its probability sources, supporting the use of complementary image and OCR-text evidence. Table 3 shows the A1 progression to our final 0.7896 macro-F1 (0.7097 F1-Hateful, 0.8696 F1-Not-Hateful), and Table 4 shows the A2 trajectory to 0.3601 macro-F1 / 0.5150 micro-F1. For A2, the selected r03 calibration reached 0.3853 on development but only 0.3099 on final test before common-label restoration, illustrating the calibration transfer problem. Among discarded development- test branches, MemeIntel (Kmainasi et al., 2025a) scored 0.2743 and Fanar/Oryx zero-shot scored 0.6444 on A1.

System	Ma-F1	Acc.	F1-H	F1-NH
Early blend	0.7539	0.774	0.6835	0.8243
Top-k calibrated	0.7794	0.808	0.7000	0.8588
Diverse top-k	0.7849	0.816	0.7032	0.8667
Surgical swap	**0.7896**	**0.820**	**0.7097**	**0.8696**

Table 3: Official A1 final-test results. H/NH = Hateful/Not Hateful. Final row is `swap_qseq_heavy_n3`.

System	Macro-F1	Micro-F1
Dev-tuned ens6 r03	0.3099	0.5130
Rare/common trim	0.3451	0.5000
Restored common + rare boost	**0.3601**	**0.5150**

Table 4: Official A2 final-test results. Best row corresponds to `hybrid_restore_common_r03_min1`.

6 Gold-Test Error Analysis

After the evaluation period, organizers released final-test gold labels, which we use only for post-submission analysis; this section was not used to build the final systems.

6.1 A1 Binary Errors

The gold distribution contains 148 *Hateful* and 352 *Not Hateful* memes; our system predicted 162 and 338 respectively. Table 6 shows the confusion matrix: 110 of 148 hateful examples are correctly identified, with 38 false negatives and 52 false positives. Errors concentrate by fine-grained label: *Incitement* examples had 10 false negatives (accuracy 0.750), and *Mocking* had 24 false negatives with 0 false positives. Conversely, *Humor* and *Sarcasm* drove more false positives (19 and 29, respectively) — the ranker appears to associate rhetorical/humorous framing with hatefulness, sometimes over-applying this cue.

6.2 A2 Multi-Label Errors

Final-test A2 labels are imbalanced and low-cardinality: the gold set has 336 single-label, 136 two-label, 25 three-label, and 3 four-label examples, while our predictions skew broader (two labels for 313 examples, one label for only 90), aiding common-class recall at the cost of subset accuracy and more false positives, especially for *Sar-*

System	Macro-F1	Acc.	Threshold
Qwen3-VL-4B	0.7511	0.764	0.510
Six Arabic encoders	0.7395	0.762	0.560
Qwen3-4B text LoRA	0.6847	0.700	0.725
Seven-model text ensemble	0.7462	**0.770**	0.550
0.60 VLM + 0.40 text	**0.7561**	0.764	0.410

Table 5: A1 ablations on the 500-example labeled development split. Each threshold is selected on this split for macro-F1. The seven-model text ensemble averages the six Arabic encoders and Qwen3-4B.

	Pred. Hateful	Pred. Not Hateful
Gold Hateful	110	38
Gold Not Hateful	52	300

Table 6: Post-release A1 confusion matrix.

casm, *Other*, and *Dehumanization*.

Table 7 reports post-release per-label statistics. Frequent categories are strongest: *Humor* F1 0.686, *Sarcasm* 0.581, *Mocking* 0.554. Rare or semantically specific categories remain hard: *Exclusion* (3 gold examples) gets 0 F1; *Dehumanization* recall is 0.381 but precision only 0.110, showing the rare-label boost recovers positives at the cost of many false positives. The largest confusions are between rhetorical categories: *Mocking*↔*Sarcasm* substitutions occur roughly 25 times each direction, and *Humor* is often replaced by *Other*, *Sarcasm*, *Mocking*, or *Slurs* — plausible given that Arabic memes frequently encode harm through irony and ridicule simultaneously.

7 Discussion

Text remains highly competitive. OCR text carried substantial signal for both subtasks; Arabic text encoders trained on social/mixed-domain Arabic were especially strong, likely because many memes contain explicit text cues even when the image adds pragmatic context.

Zero-shot VLMs were not enough. Prompted VLMs were brittle — some conservative, others inconsistent in mapping free-form output to the label space; fine-tuning as a sequence classifier was more reliable than generative prompting.

A2 macro-F1 rewards rare-label care. Since macro-F1 averages over labels, *Contempt*, *Inferiority*, and *Slurs* matter as much as *Humor* and *Sarcasm*. Early systems had higher micro-F1 but lower macro-F1 from over-predicting common labels; the best calibration restored common-label recall while preserving rare-label coverage.

Label	Sup.	Pred.	Prec.	Rec.	F1
Contempt	36	45	0.378	0.472	0.420
Dehumanization	21	73	0.110	0.381	0.170
Exclusion	3	6	0.000	0.000	0.000
Humor	217	255	0.635	0.747	0.686
Incitement	40	50	0.300	0.375	0.333
Inferiority	23	30	0.200	0.261	0.226
Mocking	103	121	0.512	0.602	0.554
Other	50	121	0.231	0.560	0.328
Sarcasm	186	317	0.461	0.785	0.581
Slurs	16	50	0.200	0.625	0.303

Table 7: Post-release A2 per-label support, predicted count, precision, recall, and F1.

The remaining gap is semantic. Remaining A2 errors involve overlapping pragmatic distinctions (sarcasm, humor, mockery, contempt) rather than simple lexical mistakes, suggesting future work should target label-dependency modeling and stronger visual cues rather than independent per-label thresholding.

Calibration mattered more than model size. Final improvements resulted from selecting optimal operating points rather than increasing model size. For A1, performance improved through thresholding, then top-k selection, and finally surgical swap. For A2, replacing global thresholds with per-label predicted-count control led to better results. Due to the shared-task submission budget, we tuned these to the appropriate operating points using aggregate CodaBench feedback rather than a held-out validation split. This calibration approach is important to consider when comparing systems.

8 Conclusion

We presented calibrated systems for Arabic harmful meme detection in ArGuard Track A. Our A1 system combines Qwen3-VL and Arabic text classifiers with top-k and surgical-swap calibration, reaching 0.7896 macro-F1. Our A2 system uses a six-model Arabic text ensemble with rare- and frequency-aware label calibration, reaching 0.3601 macro-F1. The main lesson: for small multimodal safety datasets under macro-F1, model selection alone is insufficient — probability ranking, predicted label cardinality, and targeted post-processing can decide final performance.

Limitations

Our final calibration used aggregate feedback from leaderboard submissions, which is practical under a limited submission budget but may overfit to the final test distribution and should not be read as a de-

ployable calibration procedure. VLM fine-tuning was limited to a small set of Qwen3-VL configurations; stronger backbones or joint binary/fine-grained training may help. We relied on task-provided OCR text without systematically evaluating OCR errors, visual-only cues, or dialect robustness.

Ethics Statement

This work concerns harmful and hateful content; the dataset contains offensive language and imagery by design. We used the data only for non-commercial research and did not attempt to identify private individuals. Such systems can produce false positives that suppress benign speech or false negatives that miss harmful content, especially for dialectal Arabic, sarcasm, reclaimed language, or culturally specific references. Deployment would require human review, uncertainty transparency, and monitoring for disparate impact across Arabic-speaking communities.

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